Week 3: Linux Logs, Sysmon, auditd and Snort Setup

Theory Notes, Developing Basic Understanding in SOC

Question 9. What are the key Linux log files a SOC analyst should monitor? Include a table of Log File Path | Purpose | Common Events.

These are the Linux log files that I would check first during an investigation.

Log File Path	Purpose	Common Events
/var/log/auth.log	Authentication and authorisation on Debian and Ubuntu	SSH logins, failed passwords, sudo usage, user and group changes
/var/log/syslog	General system activity and service messages	Services starting and stopping, cron jobs, network and system messages
/var/log/audit/audit.log	Detailed records from auditd	Command execution, file access, logins, privilege changes
/var/log/kern.log	Messages from the Linux kernel	Driver errors, USB device events, firewall drops, hardware problems
/var/log/apache2/access.log	Every request made to the Apache web server	IP address, URL, status code, user agent, scanning and web attacks
/var/log/apache2/error.log	Errors from the web server	Failed requests, script errors, startup problems
/var/log/secure	Does the same job as auth.log on RHEL, CentOS and Fedora	SSH logins, sudo usage, failed authentication
/var/log/wtmp	History of past logins and reboots, read with `last`	Login times, source IP, reboot history
/var/log/btmp	History of failed logins, read with `lastb`	Brute force attempts
/var/log/dpkg.log	Package installs and removals	Software installed or removed, which can show tools added by an attacker
/var/log/ufw.log	Firewall activity, if UFW logging is turned on	Blocked and allowed connections

Some commands that help

```
sudo grep "Failed password" /var/log/auth.log
sudo tail -f /var/log/auth.log
sudo lastb
sudo ausearch -m USER_LOGIN -ts recent
```

Why these logs are important

- auth.log and secure help me find brute force attacks and misuse of sudo.
- audit.log shows which commands were run and which user ran them.
- The web server logs show scanning, SQL injection and other web attacks.
- The package and kernel logs help me notice new software or unusual hardware activity.

- It is best to send all these logs to a central SIEM, because an attacker can easily delete logs on the machine itself.